(1) Data Collection

(1) What is "Data Collection"?

→ In the context of A/B testing, Data Collection means accurately records what users do, when, where and under which experimental condition (control vs. treatment)
→ If data is incomplete, inconsistent, or biased, then your experiment conclusion can be completely wrong.

"Data Collection" covers

1. Instrumentation - where in the state you fire the log and what payload you send.
2. Transport & storage - queues, ETL, warehousing, retention policies.
3. Governance - schemas, versioning, privacy controls, access logs.

Think of it as the plumbing that carries raw water (user behavior) to the treatment plant (analytics). A leak or rusty pipe contaminates the supply long before it reaches your glass.

(2) Why Data Collection is Crucial?

Imagine a clinical drug trial where half of the patients forgets to report their symptoms. Would you trust the results?

Same with online experiment:

- If clicks, purchases, or logins aren't logged properly
- If treatment log are richer or poorer than control logs.
- or if failures/successes are dropped unequally.

Then your experiment becomes scientifically invalid - no matters how good your statistical methods are.

why is it ethically critical?

- Fairness to users: A bad experiment can roll back a feature that actually helped them, or ship one that hurt.
- Privacy & consent: Over-collecting personally-identifiable data violates regulations and user trust.
- Organizational Integrity: leadership decisions, bonuses and product road-maps ride on these numbers.
  - wrong data ⇒ wrong incentives.

(3) Real-world analogy + maths.

Goal: Understand how bad logging ruins your experiments

Setup: The Loyalty Card Experiment

Two coffee shops are running an experiment:

- Control group = Shop A (old checkout flow).
- Treatment group = Shop B (new loyalty card system)

Each shop gets 1,000 customers

- At Shop A (control)
  100 customers buy something → 10% conversion rate.
- At Shop B (treatment)
  120 customers buy something → 12% conversion rate.
  (So the new loyalty card is actually helping — a +2 percentage point improvement).

✓ So, true uplift = 2%.

⚠ Problem: Bug in Data Logging

But... the logging system in shop B has a bug:

> It fails to log 40% of the successful purchases (the people who did buy)

So instead of logging 120 purchases, it log only:

$120 \times (1 - 0.4) = 72$ purchases logged.

Everything else is logged fine.

What does your A/B Test Dashboard show?

Group	True Purchases	Purchases Logged	Total Users
Control	100	100	1000
Treatment	120	72	1000

Logged Conversion

100/1000 = 10%

72/1000 = 7.2%

What's the Wrong Conclusion?

From the logged data, it looks like the new loyalty card made things worse (7.2% vs 10%) — a negative effect of -2.8%

But that's totally false.

In truth, it improved sales by +2%, but the logging error hid 40% of those successes.

Why This happens (Mathematical View)

Let's define

- $Y = 1$ if a person made a purchase
- $T = 1$ if they were in treatment
- $W = 1$ if their data were logged.

You want to estimate:

$$\Delta = E[Y|T=1] - E[Y|T=0] = 12\% - 10\% = +2\%$$

But instead you compute:

$$\bar{\Delta}_{obs} = \frac{\sum Y_i T_i W_i}{\sum T_i W_i} - \frac{\sum Y_i (1-T_i) W_i}{\sum (1-T_i) W_i}$$

$$= \frac{72}{1000} - \frac{100}{1000} = -2.8\%$$

Summary of the failure

Issue	Result
Treatment group's successes under-logged.	True lift hidden
Estimator uses only what's logged	Biased downward.
Experiment says "bad feature"	But it was actually good.

Final Insight

> Bad logging doesn't just add noise - it adds bias if the chance of logging depends on both T and Y, your A/B test result becomes actively misleading.

(4) Simple Checklist for Ethical, Reliable Data Collection.

Practice	Why it's critical
☑ Log before branching	Use same logging path in control & treatment to ensure fairness
☑ Audit Completeness	Count missing events across all arms.
☑ Standardize schemas	Avoid misinterpretation in downstream systems.
☑ Limit and hash PII	To follow GDPR / CCPA and reduce risk.
☑ Automate logging tests	Catch issues before experiment launch.

(5) Interlinked Topics.

Related Topic	Why it connects.
User identifiers (p. 123)	logging without leaking private user info.
Culture and Processes (p. 122)	Good data collection needs strong org habits.
Metrics (units)	Your metrics are only good as your raw logs.
Institutional Memory (units)	logging mistakes are often repeated unless documented.

⑥ Final Thought

> "If your thermometer is broken, no amount of math will give you the right body temperature."

Data collection is that thermometer. If it's wrong — every thing built on it becomes noise.

② Cultures and Processes 122

① What is "Cultures and Processes" in Experiments?

Culture = the shared beliefs and values of your team.
Process = the consistent actions and checkpoints your team takes.

> A good culture sets the "why"
> Good processes make the "why" happen every day.

In ethical A/B testing, it's not enough for one person to care — the entire system should be designed to catch flaws before they hurt users or mislead decision-makers.

② Real-world Example: Teaching an A/B Test

☑ In a strong culture with good processes.

1. Engineer writes a design doc.
   - States hypothesis
   - Defines metrics
   - Flags any privacy-sensitive data collected.

2. Before launch:
- Peer review board checks logging, ethics, risks.
- checklist is verified: telemetry complete, user ID anonymized, rollback ready.

3. During Experiment:
- Guardrail metrics (e.g. page crashes or revenue drops) are monitored live.
- ~~Alerts triggers~~
- Alerts triggers auto-pauses if needed.

4. After:
- A post-mortem reviews what worked or failed — even if the experiment was a success.

This builds institutional memory (see Units), avoids repeating past mistakes, and makes ethics normal, not optional.

X without culture or process

- A single engineer deploys a test late at night
- No one checks that the logging code works in both control and treatment.
- The test accidentally logs raw email addresses, breaching privacy rules.
- It misleads leadership because it drops 30% of mobile users.
- No post-mortem is done mistake happens again.

| without culture + process, ethics rely on luck and heroism — which fails at scale.

(3) Math-Backed Simulation (scroll up).

In the bar chart and table above, we simulate.

- 10,000 experiment per year
- 10% have hidden flaws
- with no checks: 1,000 bad experiment ship.
  - only ~ 80 flawed test ship
  - That's a 12.5 x improvement.

This proves that even basic review process reduce real-world risk dramatically.

(4) Simple Example of a Strong Process-

Step	Purpose	Related Concept
☑ Pre-Launch Checklist	Confirms metrics, privacy, logging	unit-7 metrics.
☑ Peer Review	Catches bias or unethical design	p.122. Data collection
☑ Guardrails setup	Pauses test if harm detected	p.116-Background
☑ Post-Mortem logging	Captures learning	Unit-2 Institutional Memory.

(5) Final Summary.

with culture & Process	without Culture & Process
Bias is caught early	Bias leaks with decisions
Privacy is protected	Data is over-collected.

Mistakes are learned from	Mistakes are repeated.
Experiment are trustworthy	Results are misleading.

(3) SIDEBAR: User Identifiers 123.

(1) Why user IDs matter?

→ An online experiment assumes every observation is independent and each user see only one bucket (control or treatment)

That hinges on a single question:

> "How do we decide who "the same user" is?"

If your identifier drifts (e.g. multiple cookies per person, cross-device mismatch, raw e-mails leaking PII), you get.

1. Cross-bucket contamination - the same person counted in both variants.
2. Biased estimates - active users over-represented, inflating variance.
3. Privacy risk - real-world identity exposed when logs are joined.

Simple

→ User identifier are how you tracks a person in an A/B test.

Examples:

- User ID (when logged in)
- Cookie ID.
- Device ID
- Hashed email
- IP address (bad idea for tracking)

The ID you choose determines the unit of randomization — and whether your experiment is valid or misleading.

2. Common identifier choices.

ID type	Pros	Cons.
Cookie	Easy, client-side	Deleted often; breaks across devices.
login/UserID	Stable, cross device	Needs account; PII if unhashed
Device ID	Works before login	One person owns many devices.
Hashed composite (e.g. salted hash of login)	Pseudonymity, GDPR-friendly	Must keep salt safe.

Rule of thumb: Prefer a single, stable, hashed user-level ID for bucketing, and use sessions IDs only for analyses within buckets.

(3) Real world Analogy / Piano Examples.

Scenario A: User-level Bucketing (correct)

- Alice logs in from 3 devices
- System uses her user_id
- She is assigned to treatment and always stays in that group

☑ Clean experiment

Scenario B: session-level Bucketing (Flawed).

- Alice visits once from her laptop (gets control)
- Visits again from phone (gets treatment)
- same person is counted in both arms.

✗ Cross-bucket contamination.

Now, say the treatment improves conversion by +10%. But Alice is 50/50 across both arms, so her real behavior cancels out. If many users do this, you measure only part of the real life.

Cross-bucket contamination

> The same person is included in both control and treatment.

Why This is Bad for Results

Let's say the treatment improves conversion rate by +10% (true effect):

- In control, Alice does not buy → recorded as 0
- In treatment, Alice does buy → recorded as 1

But now Alice is split between both arms:

- she brings the average down in control.
- she brings the average up in treatment.

Problems: Both sides include her behavior, so her impact cancels out.

Math behind it - with Simple Numbers

Let's define

- Δ = true lift = 10% improvement = Treatment - control.
- c = fraction of users who appear in both control and treatment

The observed lift becomes,

$$\Delta_{obs} = (1 - c) \cdot \Delta$$

Example:

if 40% of users ($c = 0.4$) are in both groups

$$\Delta_{obs} = (1 - 0.4) \cdot 10\% = 0.6 \cdot 10\% = 6\%$$

↓ You lose 40% of your signal.

If 100% of users were in both buckets?

$$(1 - 1) \cdot \Delta = 0$$

⇒ You'd see no difference at all - even if your features worked.

Good vs. Bad ID summary

ID Type	Result	Risk
User ID (e.g. login)	Alice always in treatments	☑ Good
session ID or Device ID	Alice goes to both control and treatment	☒ Bad

(4) links to other topics

Check	why	Interlocks
Hash & salt the PD	Pseudonymises data, reduces breach risk.	Data Collection 121
Consistent bucketing service	Stop cross-bucket leakage	Cultures & Processes p. 122
Hierarchy: user > session > page-view	Enforces correct unit of analysis	Unit 7 metrics.
Fallback PD logic (e.g., device-cookie until login then stitch)	Minimises orphan sessions	Background p. 116 (ethics)

(5) Take-away

Pick one stable, privacy-respectful user identifier, and use it every where. Get that wrong, and no amount of statistics will rescue your A/B test.